A CMF proof of the linear independence of $1, \zeta(3), \zeta(4)$

Proof development and exact certificates

6 October 2026

Main conclusion. For every $(A, B) \in \mathbb{Q}^2 \setminus \{(0,0)\}$, $A\zeta(3) + B\zeta(4)$ is irrational. The proof uses the trajectory in the handoff, with no change of initial point, direction, or product convention. For the nonvanishing argument we use all three columns of its 3×3 block; the displayed scalar candidate remains column zero. The new ingredient relative to the $\zeta(3)$ argument is a *second-derivative dual residue functional*. It supplies the missing arithmetic estimate for the $\zeta(4)$ readout.

Literature. Li Lai's preprint of 5 October 2026 states the same linear-independence conclusion as Theorem 1.6: `https://arxiv.org/abs/2610.06838`. The argument below is a CMF argument using the trajectory specified here. No novelty claim is made for the linear-independence statement itself. This document is a mathematical proof with exact arithmetic certificates; it has not yet been formalized in Lean.

I. The precise rational sequences

Use MeijerG$(5, 2, 5, 5, 1)$ with

$$s = (1, 1, 0, 0, 2; 0, 0, 0, 0, 0), \qquad v = (11, 12, -16, -17, 10; 0, 1, 2, -1, -2).$$

Write

$$W_N = M(0)M(1)\cdots M(N-1) = \begin{pmatrix} V_N & * \\ 0 & (-1)^N \end{pmatrix}, \qquad T_N = V_N^{-T}.$$

The exact matrix reconstruction and the block identity are certified in `verify_matrix.py`. Put

$$U_{34} = \begin{pmatrix} 0 & -3 & 3 \\ 2 & -2 & 0 \\ 0 & 0 & 6 \end{pmatrix}, \qquad \mathcal{R}_N = U_{34}T_N = \begin{pmatrix} p_{N0}^{(3)} & p_{N1}^{(3)} & p_{N2}^{(3)} \\ p_{N0}^{(4)} & p_{N1}^{(4)} & p_{N2}^{(4)} \\ q_{N0} & q_{N1} & q_{N2} \end{pmatrix}.$$

These entries are rational; they are not yet integer numerators and denominators. For column zero, with $t = W_N^{-T} e_0$, their ratios are exactly

$$\frac{p_{N0}^{(3)}}{q_{N0}} = \frac{t_2 - t_1}{2t_2}, \qquad \frac{p_{N0}^{(4)}}{q_{N0}} = \frac{t_0 - t_1}{3t_2}.$$

Thus the numerator row for $A\zeta(3)+B\zeta(4)$ is precisely $(2B, -3A-2B, 3A, 0)$ and the denominator row is $(0, 0, 6, 0)$. Also,

$$\det \mathcal{R}_N = 36/\det V_N \neq 0. \tag{1}$$

II. The analytic input, with the exact readouts

The technical proof reproduced in Part VII below establishes the constants

$$a = 53.28758332262020526735\ldots, \qquad b = 59.51995592483447507214\ldots$$

and the estimates

$$|q_{Nj}| \le e^{aN+o(N)}, \tag{2}$$

$$|q_{Nj}\zeta(3) - p_{Nj}^{(3)}| + |q_{Nj}\zeta(4) - p_{Nj}^{(4)}| \le e^{-bN+o(N)} \quad (j = 0, 1, 2). \tag{3}$$

Here is explicitly why its argument gives both errors. Let $\mathbf{A}_N, \mathbf{B}_N, \mathbf{C}_N$ be the $\zeta(4)$, $\zeta(3)$, and rational coefficient rows of the original rational kernel, and set $\mathbf{S}_N = \zeta(4)\mathbf{A}_N + \zeta(3)\mathbf{B}_N + \mathbf{C}_N$. The coefficientwise identity is

$$\begin{pmatrix} \mathbf{A}_N \\ \mathbf{B}_N \\ \mathbf{C}_N \end{pmatrix} = (-1)^N \begin{pmatrix} 3 & 0 & 0 \\ -2 & -2 & 0 \\ 1 & 1 & 1 \end{pmatrix} V_N.$$

At $N = 0$, direct cross products give

$$\mathbf{A}_0 \times \mathbf{S}_0 = -(0, 3, 6\zeta(3) - 3), \qquad \mathbf{B}_0 \times \mathbf{S}_0 = (-2, 2, 6\zeta(4)).$$

Consequently, as row-vector identities,

$$\zeta(3)\mathbf{q}_N - \mathbf{p}_N^{(3)} = -\frac{\mathbf{A}_N \times \mathbf{S}_N}{\det V_N}, \qquad \zeta(4)\mathbf{q}_N - \mathbf{p}_N^{(4)} = \frac{\mathbf{B}_N \times \mathbf{S}_N}{\det V_N}.$$

The bounds $\|\mathbf{A}_N\|, \|\mathbf{B}_N\| \le e^{bN+o(N)}$, $\|\mathbf{S}_N\| \le e^{-aN+o(N)}$, and $\det V_N = e^{(2b-a)N+o(N)}$ prove (3). The small-solution identification, including the complex conjugate modes, is proved in Part VII. No lower bound on any oscillating scalar error is used.

III. The second-derivative identity

The original proof uses the formal functional $\mathcal{L}_K$ defined in Part VII. For a proper rational function with integer poles at most K,

$$\mathcal{L}_K[f] = -\sum_{m>K} f'(m).$$

For polynomial parts, $\mathcal{L}_K[t^d] = B_d(K+1)$; the zeta values in partial fractions are treated as independent formal symbols Z_r. Define a second functional by

$$\mathcal{M}_K[f] := -\mathcal{L}_K[f'].$$

For proper f, this is $\sum_{m>K} f''(m)$, and explicitly

$$\mathcal{M}_K[(t-h)^{-r}] = r(r+1)\big(Z_{r+2} - H_{K-h}^{(r+2)}\big), \qquad \mathcal{M}_K[t^d] = -dB_{d-1}(K+1).$$

It satisfies the exact identities

$$\mathcal{M}_K[f(t) - f(t-1)] = -f''(K), \qquad \mathcal{M}_K[f] - \mathcal{M}_{K-1}[f] = -f''(K). \tag{4}$$

They follow directly by differentiating the argument of the already proved $\mathcal{L}_K$ identities. In particular, $\mathcal{M}_K[1] = 0$, $\mathcal{M}_K[t] = -1$, and $\mathcal{M}_K[t^2] = -(2K+1)$ are integers for integer K.

For the dual parameters $a' = 1 - a$, $b' = -b$, the initial rational kernel is $r_0(s) = (s-1)/s^2$. Its four second-functional moments are exactly

$$g_0^{[2]} = (2Z_3 - 6Z_4,\ 2Z_3,\ -1,\ 0). \tag{5}$$

The last two entries are polynomial contributions, not divergent sums. The same differential-equation telescoper as in Part VII is

$$D(-s)r(s) = g(s) - g(s-1), \qquad g(s) = \prod_j (s + b'_j) r(s).$$

Applying (4) gives the companion relation provided $g''(K) = 0$. This extra boundary condition must be checked; it does not follow just from the first-derivative argument.

Lemma 1. *For every $N \geq 1$ the dual moments defined by $\mathcal{M}$ have the exact same CMF transport as the first-functional moments, with final cutoff $K = 2N$.*

Proof. Start at $a'_0 = (0, 0, 1, 1, -1)$, $b'_0 = 0$, $K = 0$. First decrease a'_0 by $11N$, a'_1 by $12N$, and a'_4 by $10N$. Initially $s^5 r_0(s)$ has a triple zero at zero, and these steps preserve it, so $g''(0) = 0$ throughout. After these steps, the kernel has triple zeros at every integer $1, \ldots, 2N$: at 1 two new zeros join the initial factor $s - 1$, and at $2, \ldots, 2N$ all three newly inserted blocks vanish. Thus the cutoff may be changed from 0 to $2N$ by (4), with no boundary terms.

Next increase a'_2 by $16N$ and a'_3 by $17N$, increase b'_3 by N and b'_4 by $2N$, then decrease b'_1 by N and b'_2 by $2N$. The positive shifts do not remove any zero at $2N$; the first negative lower block also does not. The last block can remove one factor $s - 2N$, leaving a double zero of r, but the corresponding factor in $\prod(s + b'_j)$ restores the third zero of g. Thus $g''(2N) = 0$ at every vertex. All poles of g are below the cutoff. The companion coefficient remains positive, following $4 \to 4 + 33N \to 4 \to 4 + 3N \to 4$, and the same explicit parameter-separation checks as in Part VII show that every axis step is invertible. The gamma shift identities therefore transport all four formal moments exactly. This proves the assertion for every N; $N = 0$ is (5). □

Let J_N be the integer polynomial concomitant matrix displayed in Part VII. Its exact, generically certified transport identity applies to these moments as well. Direct multiplication at the initial point gives

$$g_0^{[2]} J_0[0\!:\!3, :]^T = (-2, 2, 6Z_4).$$

The Z_3 terms cancel identically. With

$$R_N^*(t) = \frac{(t + N + 1)_{11N} (t+1)_{11N} (t + 2N)_{8N+1}}{(t + 13N + 1)_{15N+1} (t + 14N + 1)_{15N+1}}, \quad Q_{Nj}(t) = \sum_{r=0}^{3} (J_N)_{jr} (-t - 12N - 1)^r,$$

we therefore obtain, coefficientwise for all N,

$$(-1)^N \mathcal{M}_{-10N-1}[Q_{Nj}(t) R_N^*(t)] = Z_4 q_{Nj} - p_{Nj}^{(4)}. \tag{6}$$

The cutoff is $-10N - 1$ because $s = t + 12N + 1$ and the unshifted cutoff is $2N$. The largest harmonic index in (6) is consequently $19N$, rather than the $18N$ in the first-functional formula.

The polynomial Q_{Nj} has integer coefficients and degree at most three. Since R_N^* has a monic integer denominator and degree -1, the polynomial part of $Q_{Nj} R_N^*$ has integer coefficients and degree at most two. Its $\mathcal{M}$ value is an integer by the formulas above. Thus no unaccounted polynomial or regularization term occurs in (6). Exact rational checks of all three columns at $N = 0, 1, 2, 4, 8, 12, 20$ accompany the proof; an additional exact audit verifies every one of the $72 + 144$ elementary steps at $N = 1, 2$, including the triple boundary zeros. The lemma and pairing identity, rather than these checks, prove (6) for all N.

IV. The all-prime denominator bound

Use $f_p, \ell_p, V_o, V_d, m_o, m_d, A_p, C_p, F_N, E_o, E_d$ as explicitly defined in Part VII, Section 3. These use full Legendre sums at every prime, so the following definition provides integer multipliers for every N. Define the additional original-row bound

$$B_p = \min_{\substack{0\le k\le 12N \\ m_o(k)\ge 2}} \{V_o(k) - (m_o(k)-2)\ell_p(k+17N)\}, \qquad E_{o4}(N,p) = \max\{0, v_p(F_N) - B_p - C_p\} \quad (p>3).$$

The row with coefficients of $\zeta(3)$ in the original kernel has valuation at least B_p. The cofactor expressing $(1,-1,0)T_N$ uses this row and the rational row, so E_{o4} bounds the $\zeta(4)$ numerator readout. For the dual formula put

$$h_4(k) = k - 10N - 1, \qquad L(k) = \max\{k-1, 29N+1-k\},$$

$$E_{d4}(N,p) = \max\left\{0, \max_{13N+1\le k\le 29N+1} \big((m_d(k)-1)\ell_p(L(k)) + 3\ell_p(h_4(k)) - V_d(k)\big)\right\}.$$

Indeed a partial-fraction coefficient at a pole of order m has valuation at least $V_d - (m-r)\ell_p(L)$, and the harmonic factor supplied by $\mathcal{M}$ loses at most $(r+2)\ell_p(h_4)$. Since $L \ge h_4$, their sum is bounded below by $V_d - (m-1)\ell_p(L) - 3\ell_p(h_4)$. The factors $r(r+1)$ are integers and only improve the bound. The polynomial part is integral, as proved above. This establishes the dual bound at all primes, including 2 and 3.

For $p > 3$, set

$$E_{34}(N,p) = \max\{\min(E_o, E_d),\ \min(E_{o4}, E_{d4})\},$$

and for $p = 2, 3$ set $E_{34}(N,p) = \max\{E_d, E_{d4}\}$. Finally put

$$\Delta_N = \prod_{p\le 29N+1} p^{E_{34}(N,p)} \quad (N\ge 1), \qquad \Delta_0 = 1. \tag{7}$$

The first pair of bounds clears $p_{Nj}^{(3)}$ and q_{Nj} (they are three times the corresponding previous readouts), and the second pair clears $p_{Nj}^{(4)}$. Therefore

$$\mathcal{Z}_N := \Delta_N \mathcal{R}_N \in \mathrm{Mat}_3(\mathbb{Z}) \quad \text{for every } N \ge 0. \tag{8}$$

This is an explicit all-N integrality theorem, not a definition by reduced numerical denominators.

The same floor-chamber calculation as before gives a particularly simple answer. Away from the small and exceptional primes already treated there, if e_3 denotes its exponent profile, then the new profile is

$$e_{34}(x) = e_3(x) + \mathbf{1}_{x>1/19} \qquad (x = N/p > 0). \tag{9}$$

On $x > 1$ this means that the periodic profile is increased by one. This is a finite exact claim: `verify_arithmetic.py` checks all 270 Farey-29 chambers, both initial and periodic parts, including every integer-offset state at every floor wall. The extra harmonic cutoff uses slope 10, already in the existing wall set, so no new chambers are omitted. The script uses only Python's integer and rational arithmetic.

For completeness, $E_{d4}(N,p) \le 6\ell_p(29N+1)$: at each Legendre level, $V_d \ge m_d - 4$ and the loss is at most $m_d + 2$. Together with the first-functional bound, this proves uniformly that $E_{34} \le \max(v_p(6), 6\ell_p(29N+1))$. Small primes contribute $o(N)$, and primes near chamber boundaries contribute $O(\log N)$, exactly as in the previous proof. Applying the prime number theorem to (9) therefore gives

$$\limsup_{N\to\infty} \frac{\log \Delta_N}{N} \le c_{34} = c_3 + 19 = 57.78399331966841174068\ldots < 57.784 < b. \tag{10}$$

The added 19 is the integral $\int_{1/19}^{\infty} x^{-2}\,\mathrm{d}x = 19$. There is also an independent integer-only upper certificate for the full new integral in `verify_arithmetic.py`; no fitted growth rate is used.

V. The irrationality proof

Theorem 1. *The numbers* $1, \zeta(3), \zeta(4)$ *are linearly independent over* $\mathbb{Q}$. *Consequently* $A\zeta(3) + B\zeta(4)$ *is irrational for every fixed* $(A, B) \in \mathbb{Q}^2 \setminus \{(0,0)\}$.

Proof. Write the three rows of the integer matrix $\mathcal{Z}_N$ as $\mathbf{P}_N^{(3)}, \mathbf{P}_N^{(4)}, \mathbf{Q}_N$. Equations (3) and (10) show that

$$\|\zeta(3)\mathbf{Q}_N - \mathbf{P}_N^{(3)}\| + \|\zeta(4)\mathbf{Q}_N - \mathbf{P}_N^{(4)}\| \le e^{-(b-c_{34})N+o(N)} \longrightarrow 0.$$

Suppose $h_0 + h_3\zeta(3) + h_4\zeta(4) = 0$ with integers h_i not all zero. The integer row vector

$$\mathbf{R}_N = h_3\mathbf{P}_N^{(3)} + h_4\mathbf{P}_N^{(4)} + h_0\mathbf{Q}_N = h_3(\mathbf{P}_N^{(3)} - \zeta(3)\mathbf{Q}_N) + h_4(\mathbf{P}_N^{(4)} - \zeta(4)\mathbf{Q}_N)$$

tends to zero componentwise. For all sufficiently large N, every component has absolute value less than one, so every component is zero. But

$$\det \mathcal{Z}_N = 36\Delta_N^3 / \det V_N \neq 0,$$

which forces $(h_3, h_4, h_0) = (0, 0, 0)$, a contradiction. If $A\zeta(3) + B\zeta(4)$ were rational, clearing its denominator and those of A, B would give just such a nonzero integer relation. This proves the assertion. □

The invertibility argument is the required nonvanishing step. It does not infer nonvanishing from an eigenvalue ratio or from a finite numerical test. It also explains why using all three columns of the same walk makes the proof simpler than trying to control the zeros of one column.

VI. Quantitative consequence and formalization interface

For each fixed rational pair $(A, B) \neq (0, 0)$, combine the integer numerator rows with a fixed integer multiplier clearing A, B. Equations (2)–(10) give numerator/denominator height at most $e^{(a+c_{34})N+o(N)}$ and linear-form error at most $e^{-(b-c_{34})N+o(N)}$. Reducing the fraction can only improve the resulting approximation inequality. Thus the handoff's readouts have the common lower bound

$$\liminf \delta_N(A, B) \ge \frac{b - c_{34}}{a + c_{34}} > 0.015629$$

along regular indices. The theorem ensures that their errors are nonzero. This is a common lower bound, not a claim that all pairs have the same delta.

The proof separates into the following precise formalization obligations:

1. the elementary integer-matrix lemma used in Part V;
2. the exact rational CMF product, determinant and concomitant identities;
3. the two formal residue functionals and their boundary/transport identities;
4. the explicit all-prime integer multiplier (7);
5. the finite rational floor certificate and its prime-number-theorem limit;
6. the real spectral/contour upper estimates (3).

The files verify finite algebraic and arithmetic identities with exact arithmetic. They are not presented as Lean verification of the analytic lemmas. No assertion is justified merely by sampled values of N.

VII. Complete underlying $\zeta(3)$ argument and certificates

The remainder reproduces the underlying proof so that the preceding argument has all its definitions, matrix identities, analytic estimates, and original prime-profile table in this single document. In this part, D_N denotes the first-functional multiplier, whereas the simultaneous multiplier in Parts I–VI is Δ_N. The convergence rate is a spectral calculation. The substantive

arithmetic step is to use *two formulas for the same rational numbers*, and take the better denominator bound separately at every prime. Each formula alone gives a weaker estimate. Their combination proves the bound below. The finite arithmetic certificate uses exact rational operations; a separate integer-only certificate bounds its limiting integral.

Theorem 2. *Consider the field* MeijerG$(5,2,5,5,1)$*, with coordinate order* $(a_0,\dots,a_4,b_0,\dots,b_4)$*, initial point*

$$(1,1,0,0,2;0,0,0,0,0)$$

and direction

$$(11,12,-16,-17,10;0,1,2,-1,-2).$$

Write its forward walk as

$$W_N=\begin{pmatrix} V_N & * \\ 0 & (-1)^N \end{pmatrix}, \qquad T_N=V_N^{-T}, \qquad U=\begin{pmatrix} 0 & -1 & 1 \\ 0 & 0 & 2 \end{pmatrix}.$$

Define rational numbers x_{Nj},y_{Nj} *by*

$$\begin{pmatrix} x_{N0} & x_{N1} & x_{N2} \\ y_{N0} & y_{N1} & y_{N2} \end{pmatrix}=UT_N.$$

The searched approximation is $j=0$*, equivalently*

$$\frac{(W_N^{-T})_{2,0}-(W_N^{-T})_{1,0}}{2(W_N^{-T})_{2,0}},$$

with zero-based indices and initial readout rows $(0,-1,1,0)$ *and* $(0,0,2,0)$*. There are explicit positive integers* D_N *such that all six numbers* D_Nx_{Nj},D_Ny_{Nj} *are integers, and*

$$\log D_N\le(38.783993451+o(1))N.$$

In particular, the same trajectory gives

$$\boxed{\delta>0.22521, \qquad \mu(\zeta(3))<5.4402.}$$

Here the assertion about δ *means that, for every fixed* $\delta_0<0.22521$*, every defined readout* x_{Nj}/y_{Nj}*, reduced to* p/q*, satisfies* $|\zeta(3)-p/q|\le|q|^{-1-\delta_0}$ *for all sufficiently large* N*.*

1. The analytic estimates

Put $u=(0,-1,1)$ and $v=(0,0,2)$. The estimates needed below are

$$\|vT_N\|\le\exp(aN+o(N)), \tag{11}$$

$$\|\zeta(3)vT_N-uT_N\|\le\exp(-bN+o(N)), \tag{12}$$

where

$$a=53.28758332262020526735\dots, \qquad b=59.51995592483447507214\dots.$$

These are the usual two spectral rates; they do not include denominator clearing. For completeness, Section 6 identifies the relevant solution and explains why the error estimate holds, including the two complex conjugate characteristic roots.

Let $B(n)$ be the upper left 3×3 block of the single-step matrix, so $V_N = B(0)\cdots B(N-1)$. With $H(n)=\operatorname{diag}(1,n,n^2)$,

$$H(n)B(n)H(n+1)^{-1} = A + O(n^{-1}).$$

The rational matrix A has one positive eigenvalue $\lambda_0 = e^{-a}$, and two nonreal conjugate eigenvalues of modulus e^b. The exact characteristic polynomial is

$$\chi(X) = X^3 - c_2X^2 + c_1X - c_0,$$

$$c_2 = \frac{8322072142729410690739920060773093078576512952980189707893 4559}{1484577825190137274336280508756918272}$$

$$c_1 = \frac{7313348012431074660893565965057178532750745716087164473792011003347946670031546119019415738107 57}{1464668689488842478843329157185306546523425 44}$$

$$c_0 = \frac{7662301234833431379793837786223779079446062798366756419352166299516337335112122657458985369 6}{2130520759189220333584634002331004844478493702212809977860478179}$$

Its discriminant is negative, and its unique real root is bracketed by $7.20272127327651192628226401054699695 8\cdot 10^{-24}$ and $7.20272127327651192628226401054699695 9\cdot 10^{-24}$. The identities $a=-\log\lambda_0$ and $b=\frac12\log(c_0/\lambda_0)$ give the intervals below. The certificate bounds these logarithms using an exact rational series with a rational remainder, without floating-point assumptions. The matrix and independent algebraic reconstruction are included in the proof files. In particular,

$$53.28758332262 < a < 53.28758332263, \qquad 59.51995592483 < b < 59.51995592484.$$

2. The two exact formulas

Throughout, $(t)_r = t(t+1)\cdots(t+r-1)$, with $(t)_0=1$. First set

$$R_N(t) = \frac{(t-17N)_{15N}(t-16N)_{15N}}{(t)_{10N+2}(t+N)_{10N+1}(t+2N)_{10N+1}}.$$

For $j=0,1,2$, expand by partial fractions the convergent sum

$$-\sum_{t=2N+1}^{\infty}\frac{\mathrm{d}}{\mathrm{d}t}\big((-t)^jR_N(t)\big) = A_{Nj}\zeta(4) + B_{Nj}\zeta(3) + C_{Nj}.$$

The coefficients on the right are defined by partial fractions, so the following identity is a coefficientwise identity over $\mathbb{Q}$:

$$\mathcal{C}_N := (A_{Nj}, B_{Nj}, C_{Nj})_{\text{rows}} = (-1)^N\mathcal{C}_0V_N, \qquad \mathcal{C}_0 = \begin{pmatrix} 3 & 0 & 0\\ -2 & -2 & 0\\ 1 & 1 & 1\end{pmatrix}. \tag{13}$$

No independence assertion about $\zeta(3)$ and $\zeta(4)$ is used. Also, exactly,

$$\det V_N = F_N = \frac{\prod_{r\in\mathcal{A}}(rN)!}{\prod_{r\in\mathcal{B}}(rN)!\prod_{r\in\mathcal{E}}(rN+1)!}, \tag{14}$$

where repetitions in these lists are retained:

$$\mathcal{A} = (16,17,18,15,14,17,18,19,16,15),$$
$$\mathcal{B} = (11,10,9,12,13,12,11,10,13,14),$$
$$\mathcal{E} = (10,9,8,11,12).$$

The identity follows either by the axis determinants or by checking the rational identity $\det B(n) = F_{n+1}/F_n$ and $F_0 = 1$.

The second formula is considerably simpler. Put

$$R_N^*(t) = \frac{(t+N+1)_{11N}(t+1)_{11N}(t+2N)_{8N+1}}{(t+13N+1)_{15N+1}(t+14N+1)_{15N+1}},$$
$$\kappa_N = 450N^2 + 55N + 2, \qquad S_N^*(t) = (t+\kappa_N)R_N^*(t).$$

Then, for every $N \geq 0$,

$$1 - \sum_{t=-11N}^{\infty} (S_N^*)'(t) = (-1)^N(x_{N2} - \zeta(3)y_{N2}). \tag{15}$$

The 1 is necessary: $S_N^*(t) = 1 + O(t^{-2})$ at infinity. In particular, at $N = 0$ this is $1 - 2\zeta(3)$, exactly the initial readout.

Write

$$S_N^*(t) = 1 + \sum_k \left(\frac{d_{k1}}{t+k} + \frac{d_{k2}}{(t+k)^2} \right).$$

The pole range is $13N+1 \leq k \leq 29N+1$, with double poles precisely on $14N+1 \leq k \leq 28N+1$, and $\sum_k d_{k1} = 0$. Thus

$$(-1)^{N+1} y_{N2} = 2 \sum_k d_{k2},$$
$$(-1)^N x_{N2} = 1 - \sum_k \left(d_{k1} H^{(2)}_{k-11N-1} + 2d_{k2} H^{(3)}_{k-11N-1} \right).$$

The largest harmonic index is $18N$. These are finite formulas for the actual CMF coefficients, not a second, unrelated approximation sequence.

For use with all three columns, define the integer matrix

$$J_N = \begin{pmatrix} 4N^2(13725N^2 + 2389N + 103) & -6N(275N^2 + 150N + 11) & -450N^2 - 43N + 2 & 3 \\ N(5N+1)(330N+23) & N(450N+43) & -2 & 0 \\ -(18N+1)(25N+1) & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Each readout is given by the same formal residue functional applied to $Q_{Nj}(t)R_N^*(t)$, where

$$Q_{Nj}(t) = \sum_{r=0}^{3} (J_N)_{jr}(-t - 12N - 1)^r \in \mathbb{Z}[t].$$

Its polynomial part has integer coefficients and degree at most two. Its contribution therefore has denominator dividing 6; the proper part has the same double poles and harmonic indices as above. Consequently the denominator estimate derived from this formula applies to all three columns. Section 7 proves the coefficient identities and explains the polynomial term.

3. Explicit all-N integer multipliers

Here are fully explicit bounds, including prime powers and small primes. For a prime p, let

$$f_p(m) = v_p(m!) = \sum_{r \geq 1} \left\lfloor \frac{m}{p^r} \right\rfloor, \qquad \ell_p(m) = v_p(\mathrm{lcm}(1, \ldots, m)), \qquad \ell_p(0) = 0.$$

For an integer interval $I=[a,b]$ and an integer k, define

$$G_p(k;[a,b])=\begin{cases} f_p(k-a)+f_p(b-k), & a\le k\le b,\\ f_p(b-k)-f_p(a-k-1), & k<a,\\ f_p(k-a)-f_p(k-b-1), & k>b. \end{cases}$$

This is exactly the valuation of $\prod_{r=a,\ r\neq k}^{b}(r-k)$.

For the original kernel use

$$\begin{aligned} \mathcal{I}_o &= ([0,10N+1],[N,11N],[2N,12N]),\\ m_o(k) &= \#\{I\in\mathcal{I}_o : k\in I\},\\ V_o(k) &= f_p(k+17N)+f_p(k+16N)-f_p(k+2N)-f_p(k+N)-\sum_{I\in\mathcal{I}_o}G_p(k;I),\\ A_p &= \min_{2N\le k\le 10N+1} V_o(k),\\ C_p &= \min_{0\le k\le 12N}\{V_o(k)-(m_o(k)-1)\ell_p(k+17N)-2\ell_p(k+2N)\},\\ E_o(N,p) &= \max\{0, v_p(F_N)-A_p-C_p\} \qquad (p>3). \end{aligned}$$

For the dual kernel use

$$\begin{aligned} \mathcal{I}_d &= ([13N+1,28N+1],[14N+1,29N+1]),\\ m_d(k) &= \#\{I\in\mathcal{I}_d : k\in I\},\\ V_d(k) &= f_p(k-N-1)+f_p(k-1)+f_p(k-2N)\\ &\quad -f_p(k-12N-1)-f_p(k-11N-1)-f_p(k-10N-1)-\sum_{I\in\mathcal{I}_d}G_p(k;I),\\ h(k) &= k-11N-1, \qquad L(k)=\max\{k-1,29N+1-k\},\\ E_d(N,p) &= \max\left\{v_p(6),0,\max_{13N+1\le k\le 29N+1}\big((m_d(k)-1)\ell_p(L(k))+2\ell_p(h(k))-V_d(k)\big)\right\}. \end{aligned}$$

The second term in $L(k)$ must be retained: denominator differences can be larger than $k-1$ near the left end of the double-pole interval.

For $N\ge 1$, the promised explicit integer is

$$D_N=2^{E_d(N,2)}3^{E_d(N,3)}\prod_{3<p\le 29N+1}p^{\min\{E_o(N,p),E_d(N,p)\}}, \qquad D_0=1. \tag{16}$$

Lemma 2. *For every integer $N\ge 0$ and $j=0,1,2$, the numbers D_Nx_{Nj} and D_Ny_{Nj} are integers.*

Proof. At a pole $-k$ of order m, the top partial-fraction coefficient has valuation $V_o(k)$ or $V_d(k)$ respectively. Each Taylor derivative can lose at most $\ell_p(M)$, where M bounds the absolute values of all nonzero integer factors at that pole. Thus the coefficient of $(t+k)^{-r}$ has valuation at least $V-(m-r)\ell_p(M)$. Multiplication by an integer polynomial does not weaken this statement. For the original kernel the possible factor $1/2$ from the second Taylor coefficient is harmless because this bound is used only for $p>3$.

The rational coefficient contributed by that pole also includes $H_h^{(r+1)}$, with valuation at least $-(r+1)\ell_p(h)$. Since $\ell_p(M)\ge\ell_p(h)$, its valuation is at least

$$V-(m-1)\ell_p(M)-2\ell_p(h).$$

This proves the bound C_p for every entry of the rational row of $\mathcal{C}_N$. The $\zeta(3)$ row satisfies the same weaker bound, and the $\zeta(4)$ row has valuation at least A_p. Now

$$u\mathcal{C}_0^T=(0,2,0), \qquad v\mathcal{C}_0^T=(0,0,2), \qquad \det\mathcal{C}_N=-6(-1)^NF_N.$$

Consequently the cofactors expressing x_{Nj} use the $\zeta(4)$ and rational rows, and those expressing y_{Nj} use the $\zeta(4)$ and $\zeta(3)$ rows. This gives the denominator bound $E_o(N,p)$.

For the dual formula, the same pole argument, now with $m \le 2$, proves the bound $E_d(N,p)$ for the proper part at every prime. The integer polynomial part has degree at most two, so its Bernoulli evaluation has denominator dividing 6, already included in E_d. Both formulas describe the same rational numbers, so at each prime we may take the minimum of the two bounds. No prime larger than $29N+1$ occurs. This proves the lemma; $N=0$ follows directly from U. □

4. The denominator growth calculation

We explain why this is a finite floor calculation, rather than a finite-depth experiment. For $p > \sqrt{30N+10}$ all factorial valuations in Section 3 equal $\lfloor m/p \rfloor$, and every $\ell_p(m)$ equals the indicator of $m \ge p$. Put $x = N/p$ and $y = k/p$.

Discard temporarily the primes dividing

$$\prod_{d=1}^{29} \prod_{r=-10}^{10} (dN + r).$$

For large N this is a nonzero integer of polynomial growth. The logarithmic contribution of these primes is $O(\log N)$: the exponents at the remaining large primes are uniformly bounded. Small primes contribute $o(N)$, because the explicit uniform bound is

$$E_d(N,p) \le \max\{v_p(6), 5\ell_p(29N+1)\}.$$

At each Legendre level the dual top-coefficient valuation is at least $m_d(k)-4$: in the simple-pole case there are four positive and six negative factorial floors with zero net linear argument, and in the double-pole case three positive and seven negative floors with net linear argument one. The loss is at most $m_d(k)+1$ per level. Thus small primes contribute $O(\sqrt{N}\log N)$, and the exceptional large primes have bounded exponents.

Every possible jump of a factorial floor, pole membership, or loss indicator as k varies has the form $k = cN + jp + s$, with $|s| \le 2$. The relevant sets of c are

$$\mathcal{S}_o = \{-17, -16, -2, -1, 0, 1, 2, 10, 11, 12\},$$
$$\mathcal{S}_d = \{0, 1, 2, 10, 11, 12, 13, 14, 28, 29\}.$$

The order of two distinct jumps can change only when $(c-c')x$ is an integer. Since $|c-c'| \le 29$, the Farey fractions of order 29 partition $(0,1)$ into all the required chambers. The discarded primes cover the bounded integer offsets near chamber boundaries.

On each chamber it suffices to inspect the jumps and the constant regions between them. The certificate also inspects offsets $-3, \dots, 3$ at every jump using exact rational numbers with a formal positive infinitesimal. This covers isolated endpoint states, not just open intervals. There are 270 chambers. Taking the minimum of the two resulting bounds gives a step function $e_0(x)$ for $0 < x < 1$.

For $x > 1$ the bound is periodic in x; denote it by $e_1(\{x\})$. Indeed each pole-order sector then contains a full period of residues $k \bmod p$, both loss indicators are one, and the factorial slope sums balance. The factorial determinant also has balanced slope sum $165 - 115 - 50 = 0$. The exact chamber calculation gives the table in the appendix. This calculation uses only floors of rational numbers, comparisons, and integer maxima and minima.

The prime number theorem now gives

$$\limsup_{N\to\infty} \frac{\log D_N}{N} \le c := \int_0^1 \frac{e_0(x)}{x^2}\,\mathrm{d}x + \int_0^1 e_1(x)\psi'(1+x)\,\mathrm{d}x \tag{17}$$
$$= 38.783993319668411740683361202012500637 85\ldots.$$

For example, a chamber (l, r) in the first integral contributes $e_0(1/l - 1/r)$, and its contribution from the intervals $(k + l, k + r)$, $k \geq 1$, is $e_1 \sum_{k\geq 1}((k + l)^{-1} - (k + r)^{-1})$. One first applies the prime number theorem on finitely many such intervals; the remaining x-tail is $O(1/K)$ uniformly and can then be discarded.

There is no need to trust a floating-point digamma evaluation. The first integral is exactly $9967051/272272$. Sum the rational interval terms in the second integral for $k = 1, \ldots, 4096$, rounding each upward to a multiple of 10^{-16}. Its remaining tail is at most $m/4096$, where

$$m = \int_0^1 e_1(x)\,\mathrm{d}x = \frac{1313791}{604656}.$$

This purely integer calculation gives

$$c \leq 38.7839934501989356 < 38.784. \tag{18}$$

Together with Lemma 2, this proves the arithmetic assertion in Theorem 2.

This is also the desired GCD estimate. If d_N is any integer clearing the original rational readouts and $p_{Nj}^{\mathrm{raw}} = d_N x_{Nj}$, $q_{Nj}^{\mathrm{raw}} = d_N y_{Nj}$, their least common denominator L_{Nj} divides both d_N and D_N. Therefore

$$\gcd(p_{Nj}^{\mathrm{raw}}, q_{Nj}^{\mathrm{raw}}) \geq \frac{d_N}{L_{Nj}} \geq \frac{d_N}{D_N}, \qquad \log\gcd \geq \log d_N - cN - o(N).$$

5. Passing from integer linear forms to μ

Put $P_{Nj} = D_N x_{Nj}$ and $Q_{Nj} = D_N y_{Nj}$. They are integers, and

$$|Q_{Nj}| \leq e^{(a+c)N+o(N)}, \qquad |Q_{Nj}\zeta(3) - P_{Nj}| \leq e^{-(b-c)N+o(N)}.$$

The 2×3 matrix with these three pairs as columns has rank two, because U has rank two and T_N is invertible. Thus for any rational h/k with $k > 0$, at least one column satisfies $kP_{Nj} - hQ_{Nj} \neq 0$. Choose N of size $\log k/(b - c - \varepsilon)$ so that all three linear forms have absolute value at most $1/(2k)$. For that column,

$$1 \leq |kP_{Nj} - hQ_{Nj}| \leq \tfrac{1}{2} + |Q_{Nj}|\,|k\zeta(3) - h|.$$

It follows, on letting ε decrease to zero, that

$$\mu(\zeta(3)) \leq 1 + \frac{a + c}{b - c} = \frac{a + b}{b - c}. \tag{19}$$

No lower bound on an oscillating error term is required. Likewise the approximation exponent is at least

$$\delta_* := \frac{b - c}{a + c} = 0.22521567851203716390\ldots.$$

Reducing a fraction can only improve the stated denominator-based approximation inequality. With the rigorous upper bound (18) and the spectral intervals in Section 1, the strict rounded conclusions are $\delta > 0.22521$ and $\mu(\zeta(3)) < 5.4402$.

6. Why the analytic estimates apply to this exact solution

Let $\mathbf{A}_N, \mathbf{B}_N, \mathbf{C}_N$ be the three rows of $\mathcal{C}_N$, and put $\mathbf{S}_N = \zeta(4)\mathbf{A}_N + \zeta(3)\mathbf{B}_N + \mathbf{C}_N$. The exact contour representation is

$$S_{Nj} = \frac{1}{2\pi i}\int_{2N+1/2-i\infty}^{2N+1/2+i\infty} \Big(\frac{\pi}{\sin \pi t}\Big)^2 (-t)^j R_N(t)\,\mathrm{d}t.$$

Closing to the right gives the derivative sum in Section 2, with its displayed minus sign. On this contour write $t = 2N + \frac{1}{2} + iy$. Directly from the factors,

$$|R_N(t)| \le \frac{1}{48N^4} \exp\left((15\log 210 - 10\log 24)N + \frac{29}{14}|y|\right).$$

The factor $|\pi/\sin(\pi t)|^2$ decays as $e^{-2\pi|y|}$. Consequently all three components of $\mathbf{S}_N$ are bounded by a polynomial in N times $e^{c_0 N}$, where

$$c_0 = 15\log 210 - 10\log 24 = 48.4260746572\ldots < b.$$

This excludes the two growing characteristic modes.

Here is an elementary justification even though these two roots are nonreal. In a fixed real basis the limiting matrix is a scalar of modulus e^{-a} on one coordinate and e^b times a rotation on the other two. For sufficiently small perturbations, the cone in which the latter two-coordinate norm is at least the first-coordinate modulus is forward invariant and expands at any prescribed rate slightly less than e^b. Our bound $e^{c_0 N}$ prevents entry into that cone. Outside it, the first coordinate bounds the whole vector and contracts at a rate arbitrarily close to e^{-a}. Polynomial gauges do not change exponential rates. Thus

$$\|\mathbf{S}_N\| \le e^{-aN+o(N)}.$$

Moreover $\|\mathbf{A}_N\| \le e^{bN+o(N)}$ and $\det V_N = e^{(2b-a)N+o(N)}$. The exact cross-product identity is

$$(\zeta(3)v - u)V_N^{-T} = -\frac{\mathbf{A}_N \times \mathbf{S}_N}{3\det V_N}.$$

It proves (12); the inverse-transpose product bound proves (11). These arguments give the required upper bounds without any assumption about the phase of the complex modes.

7. Coefficient identities and the boundary constant

The following elementary formalism makes the all-N claims independent of numerical recognition of zeta values. If all poles of f are integers at most K, write its polynomial and partial-fraction decomposition and define

$$\mathcal{L}_K\left[\sum_m p_m t^m + \sum_{h,r} \frac{c_{hr}}{(t-h)^r}\right] = \sum_m p_m B_m(K+1) + \sum_{h,r} r c_{hr}(Z_{r+1} - H_{K-h}^{(r+1)}).$$

Here Z_r are independent formal symbols, $B_1(x) = x - \frac{1}{2}$, and $B_m(x)$ are Bernoulli polynomials. For a proper rational function this is the formal value of $-\sum_{t>K} f'(t)$. Its exact shift identities are

$$\mathcal{L}_K[f(t) - f(t-1)] = f'(K), \qquad \mathcal{L}_K[f] - \mathcal{L}_{K-1}[f] = f'(K),$$

when the indicated functions have no pole at the boundary. For proper parts these follow by telescoping harmonic sums; for polynomials they are the identity $B_m(x+1) - B_m(x) = mx^{m-1}$.

After reflecting the first two upper gamma factors, the Meijer kernel is $(\pi/\sin\pi t)^2 r_{a,b}(t)$, with rational $r_{a,b}$. It satisfies

$$\frac{r(t)}{r(t-1)} = \frac{\prod_j (t-1+b_j)}{\prod_i (t-1+a_i)}.$$

With $D(\theta) = \prod_i(\theta - a_i + 1) - \prod_j(\theta - b_j)$ and $g(t) = \prod_j(t+b_j)r(t)$, this gives

$$D(-t)r(t) = g(t) - g(t-1).$$

Whenever $g'(K) = 0$, the four formal moments of $r, -tr, t^2r, -t^3r$ therefore satisfy exactly the fourth-order companion relation used by the CMF. The axis transformations follow directly from $\Gamma(t+1) = t\Gamma(t)$.

For the original parameters, first lower a_2 by $16N$ and a_3 by $17N$. The resulting double zeros at $1, \dots, 2N$ allow the cutoff to move from 0 to $2N$. Then raise the three positive upper parameters, raise the positive lower parameters, and finally lower the two negative lower parameters. Along this path the leading companion coefficient $5 - \sum a_i + \sum b_j$ stays positive; every axis matrix is invertible. The boundary factor in g has a double zero throughout. This proves (13) by exact transport from $N = 0$.

For the dual parameters $a' = 1 - a$, $b' = -b$, the initial rational kernel is $(s-1)/s^2$ and its four formal moments are

$$g_0 = (Z_2 - 2Z_3,\ Z_2 - 1,\ -\tfrac{1}{2},\ \tfrac{1}{3}).$$

First lower the three nonpositive upper parameters. Triple zeros at $1, \dots, N$ allow the cutoff to move to $K = N$. Then raise the two positive upper parameters, raise the positive lower parameters, and lower the negative lower parameters last. The companion coefficient remains positive and all shifts are nonsingular; $g'(N) = 0$ throughout. Thus these formal moments have the exact dual CMF transport, denoted D_N^{dual}.

The polynomial matrix J_N in Section 2 is the Lagrange concomitant of $z\prod_i(\theta + 1 - a_i) - \prod_j(\theta - b_j)$ and its formal adjoint. It obeys the exact identity

$$W_N J_N (D_N^{\mathrm{dual}})^T = J_0.$$

This can also be verified as a polynomial identity for a generic elementary upper shift and lower shift, then multiplied along the path. Its exact symbolic certificate is included with the proof files. The shift identities are first multiplied at generic parameters and only the complete transports are then specialized. The original and dual nonsingular paths described above show that both complete transports have regular specializations; a common path need not have nonsingular intermediate charts on both sides. Set $w_p = (-1, 0, 1/2, -1/3)$ and $w_q = (-2, 0, 0, 0)$. Then

$$g_0 = -w_p + Z_3 w_q + (Z_2 - 1)(1, 1, 0, 0),$$

and $(1, 1, 0, 0)$ is in the relevant kernel of J_0. Pairing and transport therefore give $g_N J_N[0:3, :]^T = Z_3 v T_N - u T_N$ coefficientwise.

At the endpoint substitute $s = t + 12N + 1$. The kernel becomes $(-1)^N R_N^*(t)$, and the first summation index becomes $t = -11N$. The third row of J_N multiplies the kernel by $-(t + \kappa_N)$. Its polynomial part is -1, whose formal value is -1. This proves (15), including its sign, normalization, lower limit and boundary constant. The other two rows give the integer polynomials Q_{Nj} used above. This proof is entirely coefficientwise.

Appendix: the exact prime-exponent profile

In the following table (l, r) is an open interval. The first exponent is used for $0 < N/p < 1$, and the second for the fractional part of $N/p > 1$. Boundary primes have already been accounted for separately.

l	r	e_0	e_1
0	$\frac{1}{18}$	0	3
$\frac{1}{18}$	$\frac{1}{17}$	1	3
$\frac{1}{17}$	$\frac{1}{16}$	2	3
$\frac{1}{16}$	$\frac{1}{15}$	3	3
$\frac{1}{15}$	$\frac{1}{13}$	4	4

l	r	e_0	e_1
$\frac{1}{13}$	$\frac{1}{12}$	3	3
$\frac{1}{12}$	$\frac{1}{11}$	2	2
$\frac{1}{11}$	$\frac{1}{10}$	1	1
$\frac{1}{10}$	$\frac{2}{19}$	0	0
$\frac{2}{19}$	$\frac{1}{8}$	1	1
$\frac{1}{8}$	$\frac{2}{15}$	2	2
$\frac{2}{15}$	$\frac{2}{13}$	3	3
$\frac{2}{13}$	$\frac{3}{17}$	2	2
$\frac{3}{17}$	$\frac{2}{11}$	3	3
$\frac{2}{11}$	$\frac{3}{13}$	2	2
$\frac{3}{13}$	$\frac{4}{17}$	1	1
$\frac{4}{17}$	$\frac{3}{11}$	2	2
$\frac{3}{11}$	$\frac{5}{18}$	1	1
$\frac{5}{18}$	$\frac{5}{17}$	2	2
$\frac{5}{17}$	$\frac{3}{10}$	3	3
$\frac{3}{10}$	$\frac{4}{11}$	2	2
$\frac{4}{11}$	$\frac{7}{19}$	1	1
$\frac{7}{19}$	$\frac{5}{13}$	2	2
$\frac{5}{13}$	$\frac{7}{18}$	1	1
$\frac{7}{18}$	$\frac{5}{12}$	2	2
$\frac{5}{12}$	$\frac{8}{19}$	1	1
$\frac{8}{19}$	$\frac{5}{11}$	2	2
$\frac{5}{11}$	$\frac{8}{17}$	1	1
$\frac{8}{17}$	$\frac{9}{19}$	2	2
$\frac{9}{19}$	$\frac{1}{2}$	3	3
$\frac{1}{2}$	$\frac{8}{15}$	2	2
$\frac{8}{15}$	$\frac{7}{13}$	3	3
$\frac{7}{13}$	$\frac{6}{11}$	2	2
$\frac{6}{11}$	$\frac{9}{16}$	1	1
$\frac{9}{16}$	$\frac{7}{12}$	2	2
$\frac{7}{12}$	$\frac{10}{17}$	1	1
$\frac{10}{17}$	$\frac{7}{11}$	2	2
$\frac{7}{11}$	$\frac{11}{17}$	1	1
$\frac{11}{17}$	$\frac{11}{16}$	2	2
$\frac{11}{16}$	$\frac{9}{13}$	3	3
$\frac{9}{13}$	$\frac{8}{11}$	2	2
$\frac{8}{11}$	$\frac{11}{15}$	1	1
$\frac{11}{15}$	$\frac{14}{19}$	2	2
$\frac{14}{19}$	$\frac{3}{4}$	3	3
$\frac{3}{4}$	$\frac{13}{16}$	2	2
$\frac{13}{16}$	$\frac{9}{11}$	3	3
$\frac{9}{11}$	$\frac{16}{19}$	2	2
$\frac{16}{19}$	$\frac{7}{8}$	3	3
$\frac{7}{8}$	$\frac{15}{17}$	2	2
$\frac{15}{17}$	$\frac{9}{10}$	3	3
$\frac{9}{10}$	$\frac{10}{11}$	2	2
$\frac{10}{11}$	$\frac{11}{12}$	1	1
$\frac{11}{12}$	$\frac{15}{16}$	0	0
$\frac{15}{16}$	$\frac{17}{18}$	2	2
$\frac{17}{18}$	$\frac{18}{19}$	3	3

l	r	e_0	e_1
$\frac{18}{19}$	1	4	4

Certificate scope. The denominator formula and the coefficient identities are all-N arguments. The 270 floor chambers and their endpoint states are checked with exact arithmetic. The upper bound $c < 38.784$ also has an integer-only certificate. Numerical trajectory checks were confined to depth 500; none is used as a substitute for an all-N identity or for the arithmetic growth argument.